

DIGITAL LOGIC DESIGN

Complements

- They are used to simplify the subtraction operation
- Two types (for each *base-r* system)
 - Diminishing radix complement ($r-1$ complement)
 - Radix complement (r complement)

For n -digit number N

$$(r^n - 1) - N \longrightarrow \mathbf{r-1 \text{ complement}}$$

$$r^n - N \longrightarrow \mathbf{r \text{ complement}}$$

9's and 10's Complements

- 9's complement of 674653
 - $999999 - 674653 = 325346$
- 9's complement of 023421
 - $999999 - 023421 = 976578$
- 10's complement of 674653
 - $325346 + 1 = 325347$
- 10's complement of 023421
 - $976578 + 1 = 976579$

1's and 2's Complements

- 1's complement of 10111001
 - $11111111 - 10111001 = 01000110$
 - Simply replace 1's and 0's
- 1's complement of 10100010
 - 01011101
- 2's complement of 10111001
 - $01000110 + 1 = 01000111$
 - Add 1 to 1's complement
- 2's complement of 10100010
 - $01011101 + 1 = 01011110$

Subtraction with Complements of Unsigned

- $M - N$

- Add M to r 's complement of N

- $Sum = M + (r^n - N) = M - N + r^n$

- If $M > N$, Sum will have an end carry r^n , discard it

- If $M < N$, Sum will not have an end carry and

- $Sum = r^n - (N - M)$ (r 's complement of $N - M$)

- So $M - N = -(\text{r's complement of } Sum)$

By using 9's Complement

When subtrahend is smaller than the minuend

General Subtraction

$$\begin{array}{r} 841 \\ - 329 \\ \hline 512 \end{array}$$

Subtraction using 9's
Complement

$$\begin{array}{r} 841 \\ + 670 \leftarrow (9's \text{ Complement of } 329) \\ \hline \textcircled{1}511 \\ + 1 \\ \hline 512 \end{array}$$

By using 9's Complement

When subtrahend is greater than the minuend

General Subtraction

$$\begin{array}{r} 841 \\ - 983 \\ \hline - 142 \end{array}$$

Subtraction using 9's
Complement

$$\begin{array}{r} 841 \\ + 016 \leftarrow \text{(9's Complement)} \\ \hline 857 \end{array}$$

(No carry indicates -ve value)

↓

-142 (9's Complement of result)

By using 10's Complement

When subtrahend is smaller than the minuend

General Subtraction

$$\begin{array}{r} 325 \\ - 641 \\ \hline - 316 \end{array}$$

Subtraction using 10's Complement

$$\begin{array}{r} 325 \\ + 359 \leftarrow \text{(10's Complement of 641)} \\ \hline 684 \leftarrow \text{(No carry indicate negative -ve value)} \\ \downarrow \\ - 316 \leftarrow \text{(10's Complement of result)} \end{array}$$

Subtraction with Complements of Unsigned

- 65438 - 5623

	65438
10's complement of 05623	<u>+94377</u>
	159815
Discard end carry 10^5	<u>-100000</u>
Answer	59815

Subtraction with Complements of Unsigned

- 5623 - 65438

	05623
10's complement of 65438	<u>+34562</u>
	40185

There is no end carry =>

-(10's complement of 40185)

-59815

Subtraction with Complements of Unsigned

- 10110010 - 10011111

	10110010
2's complement of 10011111	<u>+01100001</u>
	100010011
Discard end carry 2 ⁸	<u>-100000000</u>
Answer	000010011

Subtraction with Complements of Unsigned

- 10011111 - 10110010

	10011111
2's complement of 10110010	<u>+01001110</u>
	11101101

There is no end carry =>
-(2's complement of 11101101)
Answer = -00010011

Signed Binary Numbers

- Unsigned representation can be used for positive integers
- How about negative integers?
 - Everything must be represented in binary numbers
 - Computers cannot use – or + signs

Negative Binary Numbers

- Three different systems have been used
 - Signed magnitude
 - One's complement
 - Two's complement

NOTE: For negative numbers the sign bit is always 1, and for positive numbers it is 0 in these three systems

Signed Magnitude

- The leftmost bit is the sign bit (0 is + and 1 is -) and the remaining bits hold the absolute magnitude of the number
- Examples
 - $-47 = 1\ 0\ 1\ 0\ 1\ 1\ 1\ 1$
 - $47 = 0\ 0\ 1\ 0\ 1\ 1\ 1\ 1$

**For 8 bits, we can represent the signed integers
–128 to +127**

How about for N bits?

One's complement

- Replace each 1 by 0 and each 0 by 1
- Example (-6)
 - First represent 6 in binary format (00000110)
 - Then replace (11111001)

Two's complement

- Find one's complement
- Add 1
- Example (-6)
 - First represent 6 in binary format (00000110)
 - One's complement (11111001)
 - Two's complement (11111010)

Arithmetic Addition

- Usually represented by 2's complement

+ 5 00000101

+11 00001011

+16 00010000

+ 5 00000101

-11 11110101

-6 11111010

Discard

- 5 11111011

+11 00001011

+6 **100000110**

- 5 11111011

-11 11110101

-16 **111110000**

Questions